Torque-speed and efficiency report

SPM 6p/36s. Built on demand from the design's flux-linkage map and the drive limits set on the Torque-speed tab, with the same calculation the tab uses. The run's own report records the single solved operating point.

Key results

Peak torque	58.23 N m
Base speed	105.485 rpm
Peak power	0.7 kW
Top speed reached	443.118 rpm
Best efficiency	90.829%

Status

Qualified outputs pass a benchmark gate against a reference; experimental outputs have none yet.

Torque-speed envelope	Experimental: Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.
Efficiency map	Experimental: Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.
Map flux linkage, MTPA trajectory and inductances	Experimental: The d/q gates cover 12 operating points, not a map's full range of currents and angles, and do not bound the inductances.
Phase resistance and copper loss	Experimental: No wire is set, so the wire is sized to fill the slot to its limit: an estimate, not the resistance of a real winding. Set the wire for a resistance the phase-resistance gate covers.

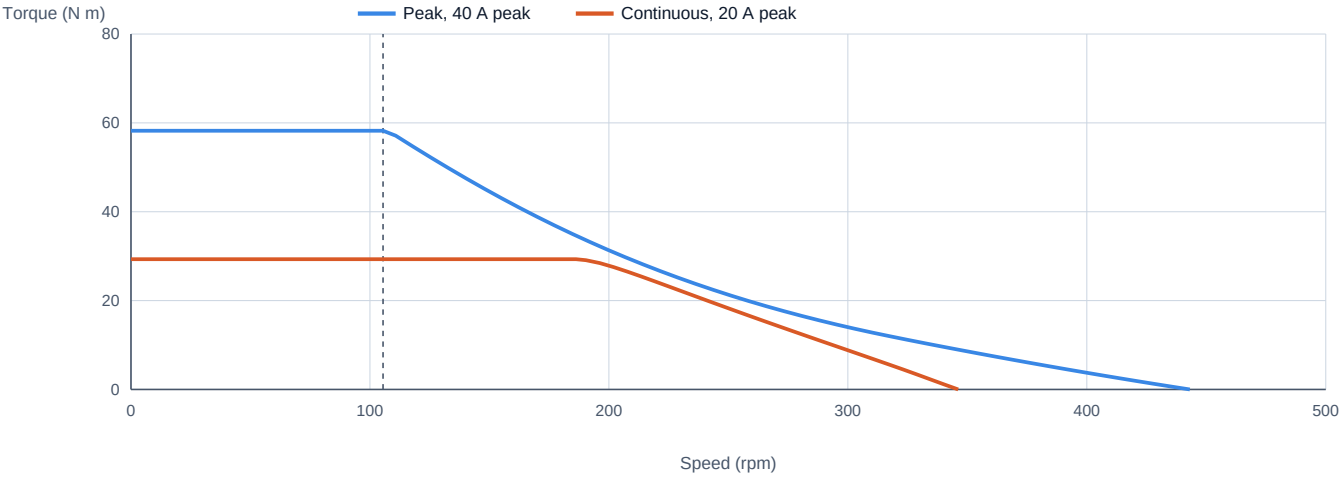
Drive inputs

Peak current	40 A peak (28.284 A rms)
Continuous current	20 A peak (14.142 A rms)
DC bus	48 V, as entered
Phase voltage limit	27.713 V peak (modulation index 1.155)
Top speed	900 rpm
Drive control	MTPA, then field weakening (least current per torque)
Phase resistance	0.412 ohm at 20 degC, experimental
Mechanical loss	0 W
Flux-linkage map	7 currents up to 40 A peak x 10 angles (0 to 90 deg), 24 rotor positions each, normal mesh, standard grid; iron loss at 300 rpm
Map scope key	5d72950c694464e1a587c353a579e8c572cbc0aabeb4b8d27bd1e25b220ec40a
coilEM	3ce46ed1ba1bcf19b422e504248917843dcf22cf

Torque and power against speed

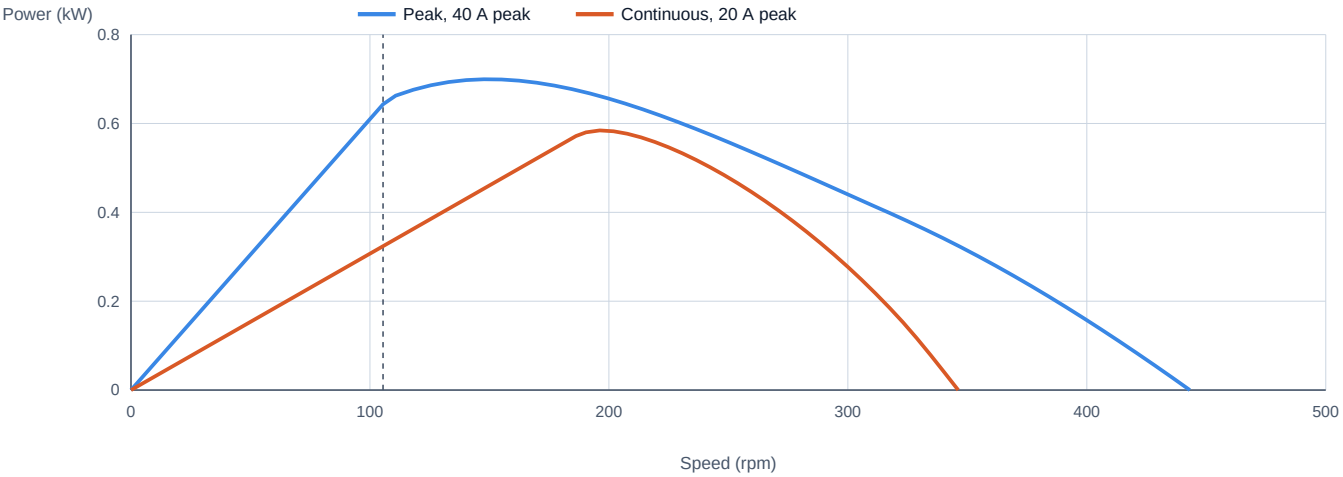
Torque against speed

Experimental



Power against speed

Experimental



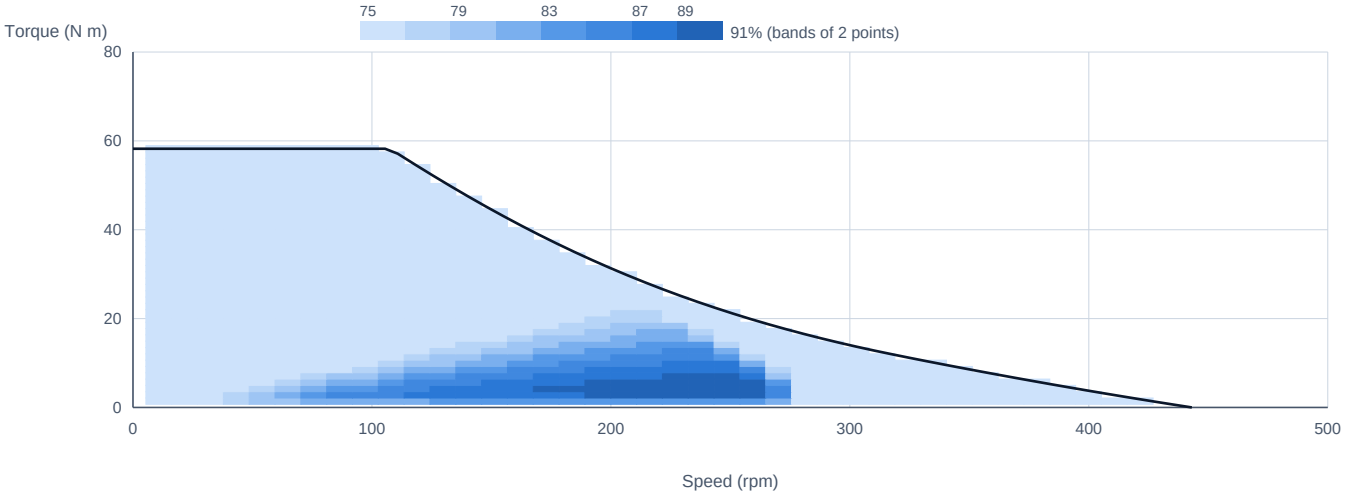
Peak curve at selected speeds (every sample is in the performance CSV)

Speed (rpm)	Torque (N m)	Power (kW)	Current (A peak)	Angle (deg)	Voltage (V peak)	Region
0	58.23	0	40	0	16.485	MTPA
44.312	58.23	0.27	40	0	21.102	MTPA
81.238	58.23	0.495	40	0	25.071	MTPA
118.165	54.625	0.676	38.579	14.4	27.713	MTPV
155.091	43.042	0.699	33.046	27.3	27.713	MTPV
199.403	31.456	0.657	31	46.2	27.713	MTPV
236.329	23.755	0.588	32.524	60.1	27.713	MTPV
280.641	16.563	0.487	36.425	71.9	27.713	MTPV
317.568	11.978	0.398	40	78.172	27.713	Field weakening
361.88	7.382	0.28	40	82.739	27.713	Field weakening
398.806	3.864	0.161	40	86.188	27.713	Field weakening
443.118	0	0	40	89.973	27.713	Field weakening

Efficiency map

Efficiency map, MTPA, then field weakening

Experimental

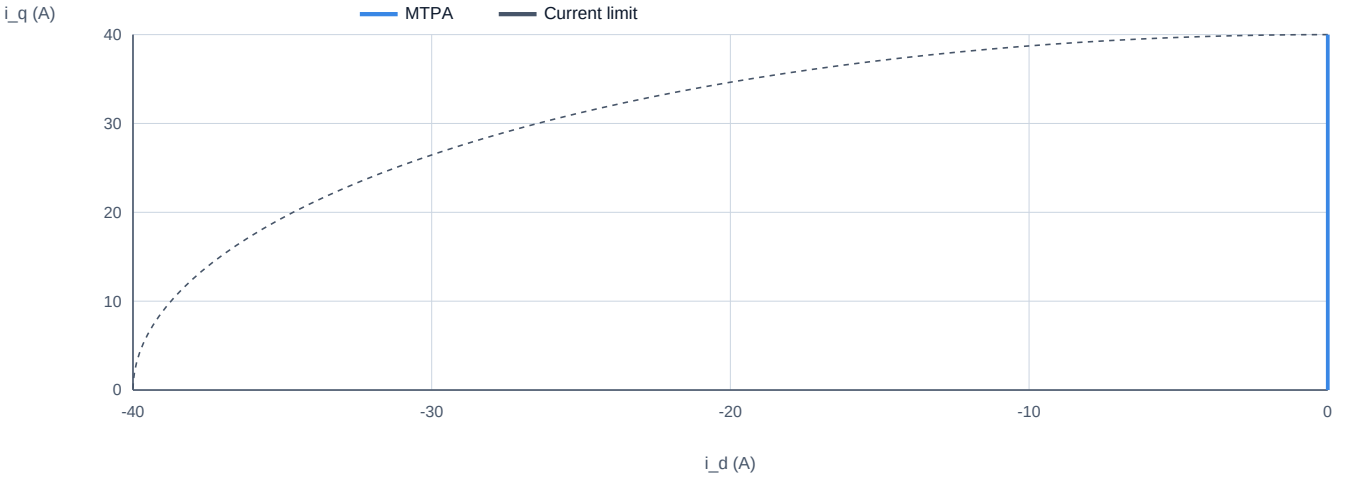


Efficiency = shaft power / input power, with shaft power = T x omega - iron loss - mechanical loss and input power = T x omega + copper loss. Cells above the peak curve are outside the drive's reach. The current and angle of every cell are in the performance CSV.

MTPA trajectory and flux linkage

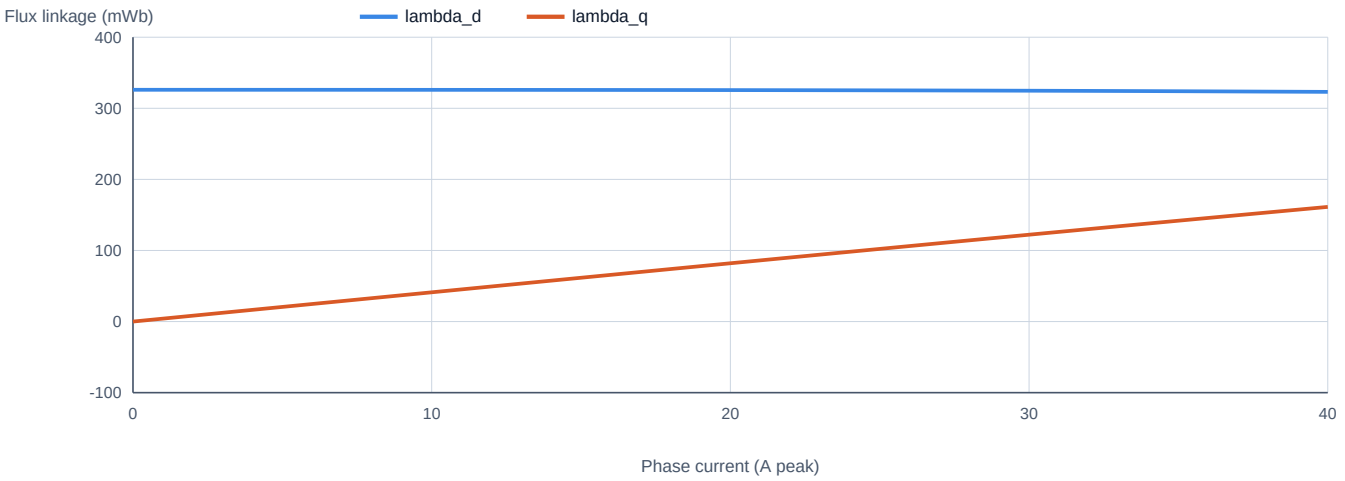
MTPA trajectory in the id-iq plane (peak A)

Experimental



Flux linkage along the MTPA trajectory

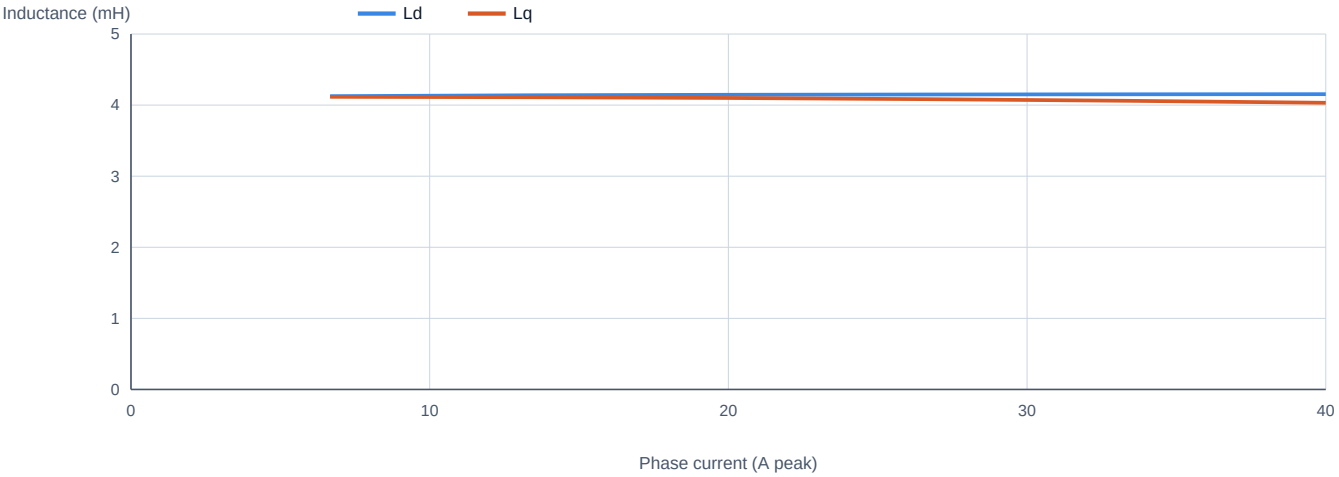
Experimental



Apparent inductance

Apparent inductance against current

Experimental



Current (A peak)	Current (A peak)	Ld (mH)	Lq (mH)
6.667	6.667	4.126	4.117
13.333	13.333	4.138	4.111
20	20	4.145	4.1
26.667	26.667	4.15	4.084
33.333	33.333	4.152	4.061
40	40	4.154	4.033

Notes

- FEM torque is 98.8-100.1% of the d/q torque across 53 map points.

- Ld from the 90 deg points, (lambda_d - lambda_pm) / i_d; Lq from the 0 deg points, lambda_q / i_q. Both are secant values and include saturation.

- The MTPA angle is read off the interpolated map, whose angles are 10 deg apart at most. At low current, where torque hardly changes with angle, it is approximate; a finer map grid gives a smoother trajectory.

- Not yet compared with measured torque-speed data. Interpolated from the flux-linkage map with sinusoidal current; inverter drops and dead time are ignored.

- Not yet compared with measured efficiency data. MTPA control: the smallest current for each torque within the voltage limit, interpolated from the flux-linkage map; torque is electromagnetic. Copper loss at the winding temperature; iron loss scaled from the map's reference speed, hysteresis with frequency and eddy current with its square.

Known limitations

- Interpolated from a flux-linkage map (Catmull-Rom over current and current angle); nothing is extrapolated beyond its currents and angles. Not yet compared with measured data.

- Steady state with sinusoidal current. Inverter voltage drops, dead time and the voltage behind the iron-loss resistance are left out.

- Torque is electromagnetic. Shaft output subtracts iron loss and the mechanical loss entered for the design.

- Iron loss is scaled from the map's reference speed: hysteresis with frequency, eddy current with its square.